

Summary of “AI on the Edge of Space: Securing Space Superiority and Avoiding Surprise in Orbit”

The U.S. Space Force faces growing threats from near-peer adversaries capable of targeting U.S. satellites, underscoring the need for enhanced space control capabilities. This paper examines how artificial intelligence can augment space domain awareness (SDA) and orbital warfare functions to help avoid operational surprise in orbit. Integrating AI, both on ground systems and onboard satellites, is essential to accelerating decision-making, enhancing satellite survivability, and maintaining domain knowledge in an increasingly contested environment.

This analysis reviews emerging AI applications upon two space mission areas and proposes additional areas for research. For the SDA mission area, it highlights the power of neural networks and explainable AI tools, such as Local Interpretable Model-Agnostic Explanations (LIME), to accelerate space object detection and improve sensor tasking efficiency. For the orbital warfare mission area, it explores how onboard AI agents can be applied to autonomously manage engagements through rendezvous and proximity operations (RPOs), optimize other satellite subsystems, and enable responsive payload tasking—within the constraints of satellite power and compute limitations. These findings are informed by recently published technical papers and defense policy documents.

The paper concludes with recommendations for responsible AI adoption into the above mission areas. These include the immediate adoption of some more mature SDA models, and procuring upgradeable satellite systems with sufficient onboard compute. This paper also recommends key policy considerations, such as defining boundaries for on-orbit autonomy, and establishing rigorous test and evaluation protocols to ensure transparent and auditable AI. In aggregate, implementing all or some of these efforts could significantly increase satellite survivability, and create opportunities to gain an algorithmically informed advantage to secure space superiority.

For more information:

- Download the report: <https://cset.georgetown.edu/publication/ai-on-the-edge-of-space/>
- Contact us: cset@georgetown.edu